Work Manager

G J Garwood

A real-time control system has been developed and deployed nationally to support BT's work management programme. This paper traces the history, system architecture, development, deployment and service aspects of this very large programme. Many issues covering technical, procedural and operational aspects had to be tackled and overcome.

1. Introduction to Work Manager

BT employs a large workforce to install and repair telecommunications equipment both at customers' premises and within its own telephone exchanges and transmission stations. Many are mobile and the challenge is to be able to assign all required work (upwards of 150 000 jobs per day) in the most effective manner to all available people (20 000).

The work management programme provides BT with the ability to assign work to people in an optimal manner and measure the performance. This involves understanding, defining and implementing work-flow processes in both the reactive repair and the provision domains, across both the customer access and core networks. Further, it is responsible for defining any development required by operational support systems (OSS) to support these processes.

Work Manager is a collection of interconnecting computer controlled systems that support this programme by automatically collecting, allocating and distributing work to BT's field technicians, while providing management information to monitor the progress and measure the effectiveness of the task. The central computer is known as the Work Manager System (WMS).

2. Historical background

In the mid-1980s, as part of the expert system/artificial in telligence (AI) research programme [1], investigations at BT Laboratories (BTL) were looking at how AI technology could be applied to BT's operations. The focus of attention was in the development of an allocation algorithm which could match people to work. Using such languages as Prolog and Poplog, demonstrator systems were produced and underwent trials with the co-operation of BT's operational divisions [2—4]. This very early work in such domains as appointment/diary management, fault analysis, job allocation to people in both repair and provision domains together with a proposal on how to achieve repair service centre logistics [5], laid the technical foundation for today's work management programme. As a result, BT was able to demonstrate (at Telecom 87 (Geneva)) a knowledge-based management system with the ability to deliver a work allocation system matching available people to required work.

By 1988 a personal computer (PC) single-user production system known as the district control support system (DCSS) [6], was launched in one area, with some 40 systems delivered nationally by 1989. This allowed each operational unit to manage out-of-hours call-out by matching the call-out condition to the most appropriate person, based on required skill, availability and location.

Recognising the success and importance of DCSS to BT, a major development programme was started in 1989 to deliver the network operations management system (NOMS2) for the core network domain. This multi-user application, supported network operations units (NOUs) and network field units (NFUs) operations within the work domain by assisting control officers to manage work and people and manual allocation.

Being a much larger application, one system per NOU was deployed utilising open system platforms based upon the Unix operating system and the Oracle database manager.

The work management programme (WMP) was born during the late 1980s, based on the early success of DCSS and the development work of NOMS2. This programme focused on the development of Work Manager as a company-wide solution for all work management related tasks. The key driver was to automate many of the previous manual allocation tasks. Figure 1 traces this development evolution path indicating that currently there are systems not only for both the customer access and core network areas but also for the national business communications (NBC) domain to support their workforce within the large business systems community.

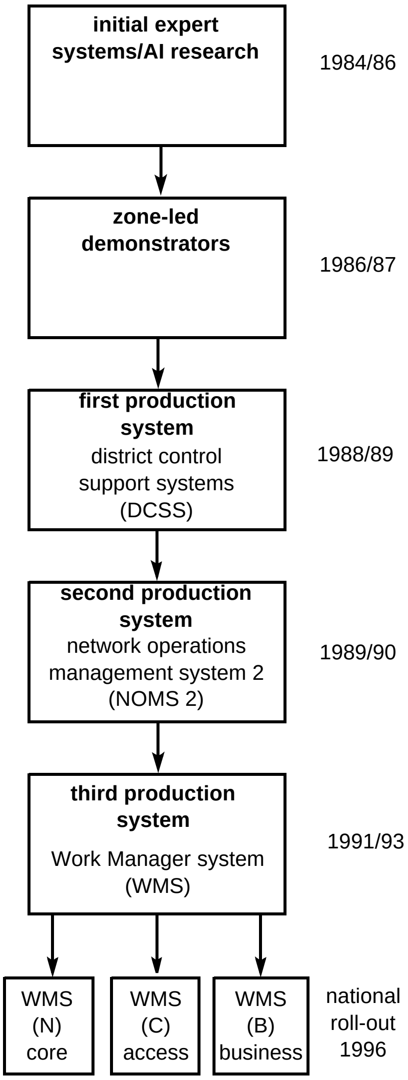

Fig 1 Historical development of Work Manager.

3. Requirements

The work management programme had to define and develop not only a system but also the operational processes to support the business requirements for the next 10 years, laying the foundation for BT's operations for the 21st century.

The three principal aims were to:

- achieve significant improvements in operational productivity by reducing unit costs in the domains of reactive repair and provision,
- maintain the required quality of service with customers,
- provide a singular common approach to work management to form the basis for all future applications.

These represented a significant challenge in that the first two points are orthogonal to each other, since one can generally only be achieved at the expense of the other, but improving both together is another matter. Achieving the third aim requires a very flexible system architecture that could both be adapted to the many related work management tasks, and evolve as the company's organisation changes.

To encapsulate the above aims into the programme's main objectives, the following simple statements were originally set and are still appropriate today:

- get the right person,
- in the right place,
- at the right time,
- with the right equipment/information,
- in an optimum and cost-efficient manner.

Thus work management in business terms is one of the key enablers.

The programme had to recognise and align itself with other strategic initiatives such as the network administration implementation programme (NAIP) [7] and the strategic systems plan (SSP) [8] and more recently with the direction of the 'Breakout' programme, all three being company-wide programmes to improve the effectiveness of BT.

The key requirement to provide a company-wide solution, which would allow the move away from manual allocation to an automatic facility, represented a significant cultural shift in that it would allocate work on a 'just-in-time basis'. Thus individuals would no longer be able to determine and plan their own daily work pattern. This change is as significant as that which took place when the company moved towards automatic telephone exchanges and away from manual boards.

In technical terms, the solution had to automatically collect jobs from the appropriate systems, allocate the work and despatch it to field technicians via a range of terminals. Figure 2 gives a simple schematic overview of the arrangement. It had to monitor progress and effectiveness, and close the work on completion of the jobs. In total the overall solution had to handle 150 000 plus jobs each working day, supporting 20 000 field technicians plus 2000

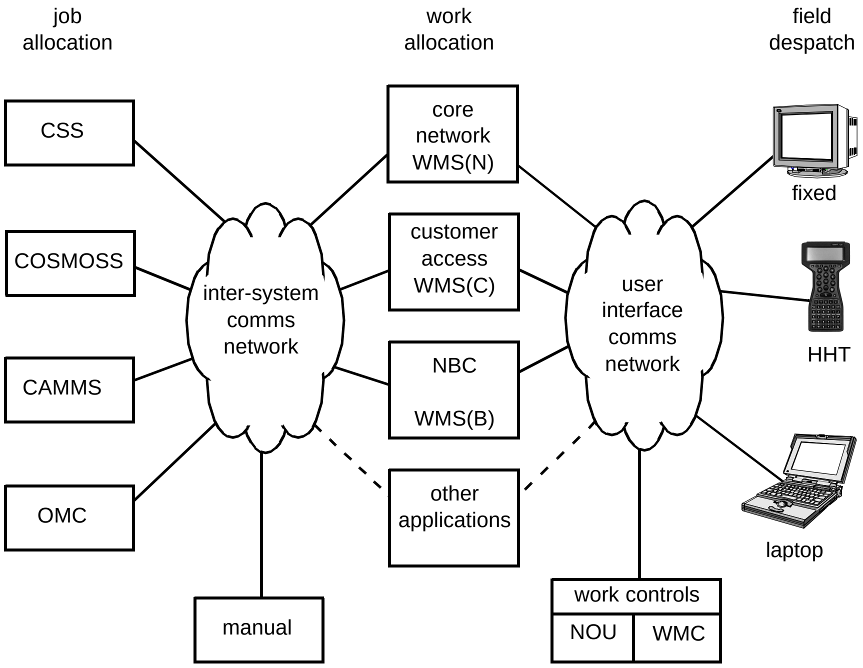

Fig 2 Schematic overview of the work management programme.

control officers. Consequently it had to be flexible, reliable, serviceable, responsive and enduring.

To support the above, a multi-million pound programme, known as the work management programme (WMP) [9], was launched in 1989. This was led and managed by people at the centre of BT's operations in conjunction with access and core field users, who defined the requirements to enable BT's development unit to design and deliver the technical solution.

4. System architecture

A national work management system had to accommodate both the core and access domains across both the reactive repair and provision workforce. Given the range and size of the tasks and current computing technology available, a single physical system was neither practical nor desirable. The proposed architectural solution was to develop a logical structure (see Fig 3) with the capability of being realised as a number of separate and independent physical systems.

At the centre of the structure would be the Work Manager System (WMS) itself. This would be interconnected to other existing support systems (e.g. customer service system (CSS) [10]), where appropriate, along with any new systems that would need to be developed.

The design objective for WMS was to realise a single logical solution (see Fig 4), but to meet operational requirements this would be available in two variants to be known as WMS(N) and WMS(C).

In the core domain, which already used NOMS2 to support the NOU and NFU operations, WMS(N) would be the direct replacement [11], utilising the existing infrastructure.

In the access domain, where no such system existed, WMS(C) and the required supporting infrastructure had to be provided anew, embracing:

- the existing CSS,
- WMS itself,
- a field access interface system (FAIS) to provide access to/from WMS from field terminals,
- a hand-held terminal (HHT) for use by the mobile access workforce.

The total work management solution, as given in Fig 3, shows the size and complexity of the arrangement. It utilises the company's main computer network for interconnection to all 29 CSS databases, the existing terminal network for local communication to PCs, and the PSTN for HHTs. In the future WMS will be able to be connected to other systems such as COSMOSS, CAMSS and OMC.

To provide the initial national coverage, the design was required to provide for 10 WMS(N) and 16 WMS(C) systems, but the design provided the flexibility for additional WMSs to be accommodated, as demonstrated by the later addition of WMS(B) for the business domain (see also section 6).

Within the architecture three basic tasks are undertaken — job collection, work allocation and field despatch.

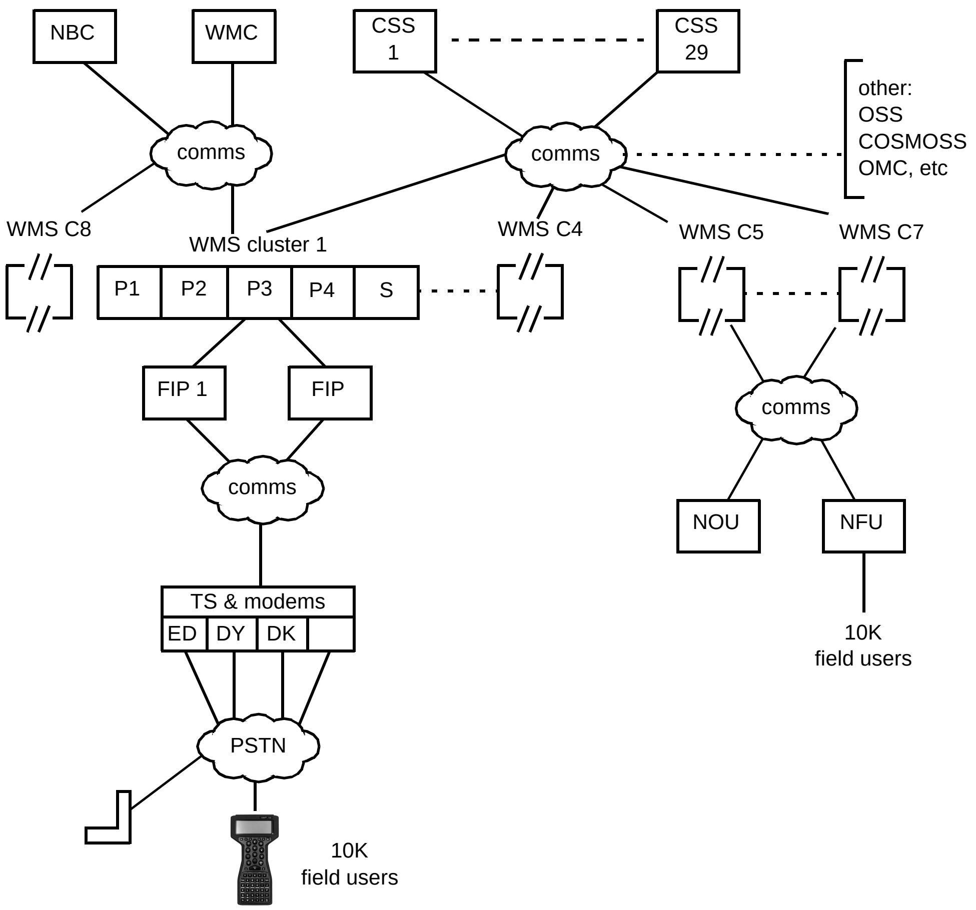

Fig 3 Work Manager structure.

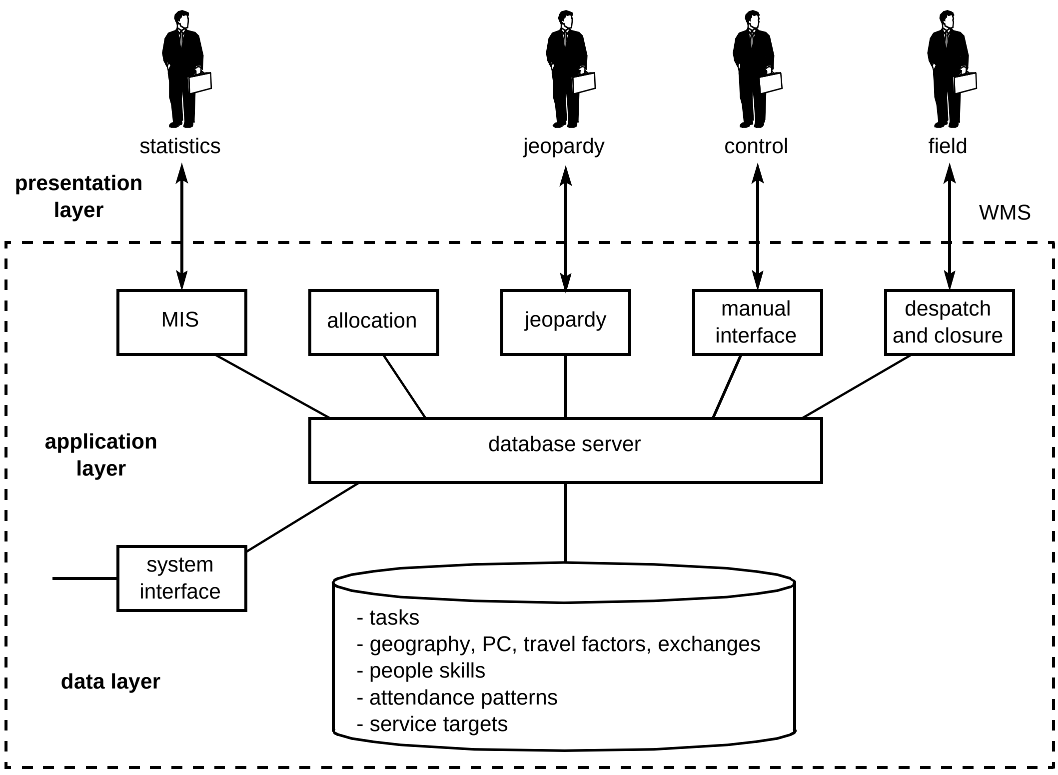

Fig 4 Work Manager logical design.

4.1 Job collection

The company collects work in the form of jobs to be undertaken. These can be held on paper, but are generally stored on a wide range of computer systems, such as CSS. WMS scans the support systems' output files and transfers details periodically via a bespoke interface. A manual interface has also been provided to allow those jobs not collected automatically to be entered. The frequency of transfers is variable but a suitable time interval is every five minutes.

Initially a manual link was used to transfer job closure information from WMS, but this has now been superseded using the system interconnect service shell (SISS) technology [12].

A generic interface has now been designed to overcome the need for further bespoke interfaces.

4.2 Work allocation

The central system for work management is the work manager system (WMS) itself. Indeed it could be described as undertaking the combined heart and brain function, being a real-time allocation control system rather than a support system. To function, it relies on its own intelligence rather than that given by manual input.

The key functions are :

- to automatically collect jobs from other support systems such as CSS,
- to automatically allocate work to the most appropriate technician in a business optimum manner,
- to automate and deliver the despatch and closure process on demand,
- to maintain full visibility of the status of both jobs and workforce,
- to generate on-demand, comprehensive management information statistics (MIS).

The functional outline is shown in Fig 5. The system is required to support residential and business repair and provision across customer access and core network workforces. The resultant work-load is highly variable. Some activities are known in advance and can thus be planned and scheduled, but others are unsolicited (e.g. customer fault reports) and have to be dealt with as they arrive.

Having collected the jobs from the various inputs, WMS uses its real-time algorithm (RTA) to match all the required work to be undertaken to all the available technicians within business imperatives. This task is a continuous process, as all the allocations have to be constantly re-adjusted as changes to real-time events occur.

The various attributes that have to be matched are listed in Table 1.

Table 1 Attributes to be matched by the real-time algorithm.

Job	customer's number type of work target completion time location skill level required appointment time type of equipment estimated duration number of people required
People	identification current work-load overtime availability start, current and finish location skill available preferred work planned absences number of people available
Business	productivity targets geographic information priorities QOS and SLA targets travel time job bundling allowed overtime risk of failure job dependencies

The RTA balances the many real and virtual costs for each job and produces the 'best' overall cost when assigning the work to each technician. Further, account is taken of those people who are about to finish, and could handle the job in the remaining allotted time. However, if unforeseen events prevent a person completing their current job on time then their next assigned job will be re-allocated to an alternative technician and a new job found for the original person when their task is completed.

For further information on the RTA, see Laithwaite [13].

Each WMS was initially designed to handle up to 10 000 jobs daily for up to 1200 technicians, but with improved computing technology this will extend to 2000 technicians, allowing even better allocations to be made.

4.3 Field despatch

Once the work has been allocated, it is held in the pre-assigned state for each technician. The normal arrangement

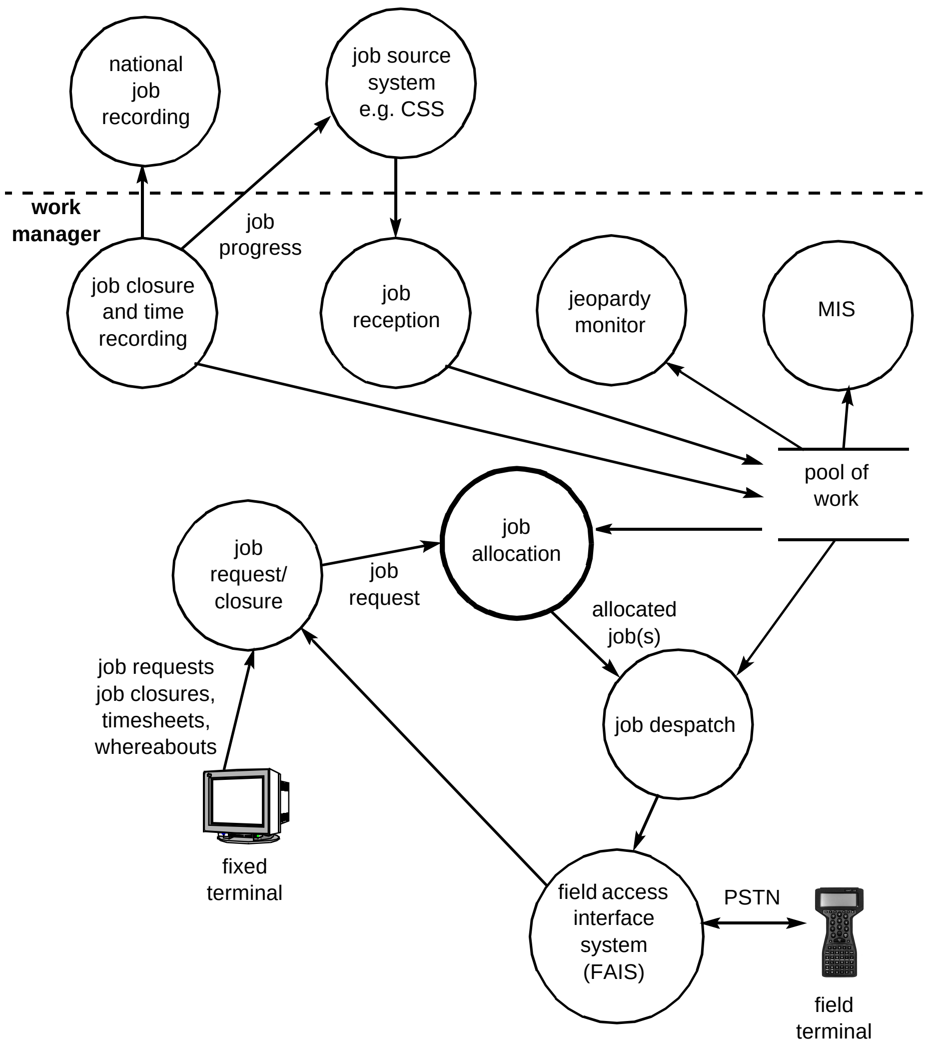

Fig 5 Work allocation and despatch system.

is to wait for the technician to call the system (log-on); this can be achieved in a number of ways.

In the core network domain the existing fixed terminal infrastructure network is utilised, where technicians in locations such as exchanges or transmission stations use either dumb terminals or PCs connected directly to WMS(N). In addition a facility to page the required person (for urgent work) is available.

In the customer access domain where the vast majority of technicians are mobile, visiting customer premises in vans over a large geographical area, technicians are equipped with a hand-held terminal (HHT). This device, which is in fact a miniature PC, calls the front-end processor (FAIS) via the PSTN. A connection is made either directly via either a telephone socket, a telephone acoustic coupler or a mobile (GSM) connection, using a non-metering pre-allocated telephone number. The most recent development is the introduction of a LapTop PC as the technician's tool which is likely to progressively replace the HHT.

The task undertaken by the FAIS is to manage the load, monitor, and control the HHT. Further it manages the access rights, security aspects, line protocols and exchange of data.

Once the link is established, details of current work, closure information and progress is uploaded. The option to undertake remote line testing is now available. At this time details of the next assigned job are downloaded. At the beginning of the working day when the technician first logs on (which can be done from home), their first assignment is allocated. At the end of the working day or when no further suitable work is available, the technician can be advised to ring control or return home or to base. Work Manager is fully integrated with the customer service improvement programme (CSIP).

To complete the work cycle, WMS(C) provides a store-and-forward function which updates CSS. This update can cover not only work-related information but also job times, allowing pay information to be interchanged with the job recording programme.

4.4 User interface

WMS has been equipped with a wide range of user tools. The zones can select which transactions are used to control the flow of work and how that work is monitored. Extensive use is made of drop-down menus with defined fields that have to be filled in.

Data build tools also exist to allow users to create the working environment they require. Parameters on people, jobs, locations, geography constraints and travel factors can all be created, modified or deleted.

In those customer access domains where WMS(C) is utilised, Work Manager controls (WMC) have been established to monitor and control the whole operation. For core network domains, the already existing NOUs have accommodated the WMC function.

4.5 Jeopardy monitoring

The key to meeting customer and business targets is jeopardy management. This is undertaken in real time to provide a continuous picture of the progress of the total work and the availability of the resource to undertake that work. Early warning of any work likely to fail its agreed completion target is flagged in sufficient time to allow alternative action to be undertaken. Indeed vital work can be reassigned to safeguard customer agreements and contracts.

4.6 Management information

Management information statistics (MIS) is a module within Work Manager and provides a raft of real-time data and trend information.

The main tool available within the standard software has restricted access to safeguard system performance and generates wide-ranging statistics on anything from productivity performance to quality of service. Statistics can be measured on a zonal, patch or even individual basis. Examples range from the number and types of jobs done each working day to management targets and failure reports.

To allow complete flexibility on a zone-by-zone basis, privileged individual users, under controlled access, have the ability to create their own statistic requirement.

This MIS is a powerful management tool, able to provide virtually any type of report, either as a spread sheet or directly on screen. It can be run in background or on demand, providing up-to-date real-time information.

4.7 WMS(C) and WMS(N)

Principally, work manager was designed as a single logical system, but to support both core and access domains with their widely different roles and management organisations, two variants have been delivered. In reality it is a single delivery with each variant being separately configured at installation. At this stage approximately 70% of the total software is common to both variants.

Nevertheless, the design is such that, as the operational requirement migrates towards a single common workforce, the software can be modified to produce one single common Work Manager system.

4.8 Fast-track solutions

Given the size of the development programme and the complex nature of some operational requirement, the WMP utilised a fast-track process. Under this arrangement individual zones could specify outline requirements which a small team translated into a PC-based solution, within a maximum of 90 days at low cost.

This had the advantage of delivering an urgent requirement quickly, provided it was self-contained and could extract its information via currently available interfaces. The disadvantage is that they are outside the main system developments, and are therefore vulnerable to any change. This imposes a significant change management and testing overhead. Although a number of fast-track solutions have been delivered, future use of this arrangement needs to consider the whole-life costs.

Nevertheless, the great benefit of this approach is to bring clarity to user requirements and ascertain benefits which, in the fullness of time, can be embedded into the main application software.

5. Development

As stated earlier, the development of NOMS2 was based on the open systems approach. This strategic decision was vital in that it allowed many parts of NOMS2 to be reused in the development of WMS. However, the most fundamental benefit to accrue from this decision was the ability to be able to port software from one hardware platform to another.

To support the open approach, the implementation was undertaken using the standard software languages C and C++ running under the Unix operating system and employing the Oracle relational database. The application software, which is large (over 2 million lines of actual source code), has been developed on a different platform to that employed in the field. Each day a port takes place to help to ensure constant portability; WMS can be ported to any large

Unix platform. During the course of NOMS2 and WMS development no less than five major Unix platforms have been used.

This programme was one of the very first large systems to have its production application software ported to a completely different manufacturer's platform. The saving in procurement by having this ability was very significant.

Extensive use of software development tools, such as Software through Pictures (StP), greatly enhanced the development in both quantitative and qualitative terms.

6. Deployment

WMS was designed as a system for national deployment. Its design had to be flexible to evolve with the changing organisation of BT.

For example, at an earlier restructuring of BT's organisation into local and national networks, the WMS was easily realigned to fit into this new framework. The technology and cost did not, however, permit the deployment of the full worker/standby arrangement; nevertheless it did provide a level of fall-back in the event of machine failure. The programme also provided a complete cluster at one site with the dual purpose of being a reference centre and a fallback in the event of a site disaster.

After a successful pilot using the WMS(C) configuration, the NBC domain is now being provided with its own cluster to support its entire BT operations. This clearly demonstrates the flexibility of the original design and capability in that a new business domain can be accommodated without the need to build a new system from scratch. Not only is WMS portable but it is also reusable.

At present eight clusters in addition to the fall-back centre are deployed.

7. Service and support

WMS has been designed to deliver a service availability of 99.8% over the working day (06.00 — 22.00) across all systems and recently independent sources have measured the WMS(C)s at 99.84%. Unfortunately, for organisational reasons it was not possible to measure the WMS(N) in the same manner but the indications were that this achieved similar performance.

During 1995/96 BT progressively introduced its open system management framework (OSMF) for its computer systems. Work Manager was modified to fit into this structure. Performance agents are now embedded with both WMS and FAIS to enable a central control at one computer centre to monitor the state of all systems at all operational sites. Progress on implementing the on-line back-up tool (known as ADSM) and the up-grade transfer facility (known as Open X-FER) is continuing.

In the past, Work Manager was supported by two different organisations from two dissimilar computer centres. Subsequently the management of both service and support has been brought under one single control and the processes involved and its implementation are now acting as a role model as other systems are being integrated into BT's central computing services.

Providing an acceptable level of service is paramount to a successful programme; WMS has made good progress but more can be achieved. Once the systems fully conform to all aspects of OSMF the business can expect even higher levels of service and availability. This approach must now be applied to the communications infrastructure, which at present is the weakest aspect of WM service.

8. Programme management

The work management programme is one of the largest and most complex undertaken by BT. It has brought together many parts of the company each with its own distinctive roles and responsibilities. Over the years the management organisation of the programme has evolved to its current structure as shown in Fig 6.

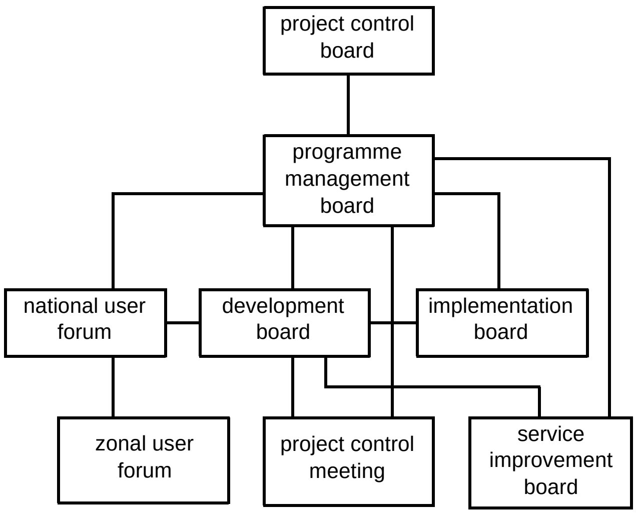

Fig 6 Management structure.

At the most senior level the project control board (PCB) steers the direction of the programme, while the programme is governed by the programme management board (PMB). At the appropriate management level, both boards are represented by field users from the access, core and business domains, the development organisations and the customer central project team.

At the lower levels the national and zonal user forums debate and scope user requirements, which are then specified by WMP development board in conjunction with the development and operational units. The implementation board manages the delivery of the function to operational systems via pilot and then roll-out schemes. The service improvement board plays the vital role of monitoring and improving service.

The key aspect of the structure is that all parts of the company are represented and contribute to the end solution, with the end users providing the most vital role.

9. Potential

The Work Manager technical vision is to provide the company's operational managers with a technically independent solution which gives them the ability to organise and manage all their engineering workforce as a single integrated unit.

The initial architecture of WM has provided a sound foundation on which to build a company-wide solution with sufficient flexibility and potential to evolve towards the above vision. By the year 2000, with the migration to a three-tier architecture, WMS could evolve from its present singular logical design to a multi-application structure which would support all aspects of work management, including the proposed access operations unit and its NetMaster system (see Fig 7). Such an architecture, which anticipates pragmatic enhancement of computing technology, will deliver many aspects of work management covering:

- work allocation,
- work scheduling,
- work collection and distribution by standard interfaces,
- network information ,
- fault diagnosis and location ,
- people information ,
- resource management.

This virtual machine would have sufficient capability to support the largest current operational zone, negating boundary conditions, and thus enabling the company to move towards a single managed workforce (sometimes referred to as a multi-domain workforce), within a very large geographical area, covering many types of work, e.g. reactive repair, provision and capital. The potential for

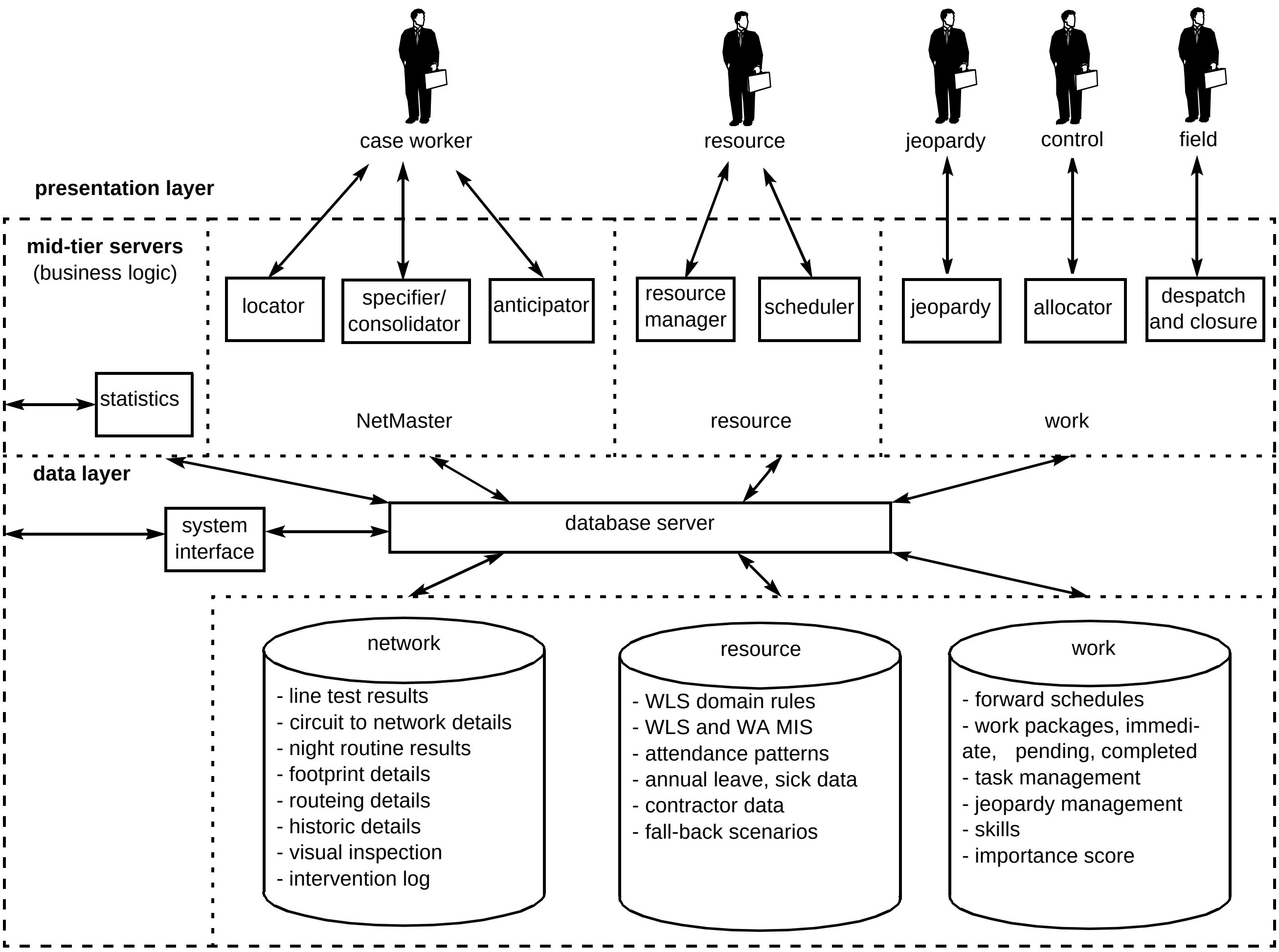

Fig 7 Work management logical design — potential vision.

Work Manager as an integral part of work management is enormous. Indeed it provides the foundation to manage any type of workforce.

10. Lessons

Any large programme will identify many lessons through its life cycle and WMP is no exception. The following sections describe some of the key lessons covering technical, procedural and operational aspects.

10.1 Technical

- Assess technical complexity and never underestimate the task. Emulating real-world attributes is difficult. It will not be achieved first time and designers must be prepared to experiment and test in operational situations. Allow much more time to understand the requirements and where possible use fast-track systems to demonstrate the proposed solution.
- Recognise that a national system is a much larger version of a demonstrator, and attention to scale is vital. Consider carefully scalability issues across all hardware and software domains.
- Service operability and availability can easily be underestimated. These attributes must be designed in at the beginning. Supporting standard service arrangements is vital.
- Recognise the capability and limitations of the communications infrastructure — this aspect is not in the direct line of control of the programme which has to influence its evolution rather than direct it.
- Use state-of-the-art and proven technology, but ensure that the future path is evolutionary.

10.2 Procedural

- All technical solutions must support company processes. Do not expect technology to provide the process. Understand the business requirement being addressed and explain proposed solutions to clarify the need. A simple demonstrator can help enormously.
- Establish a company champion at very senior level to support the programme.
- Establish team working across field users, central management and developers. Consult, advise and liaise on all issues at all levels. A sound programme management structure is vital.
- Develop the programme vision, identify the benefits/ costs and explain these in both business terms and local needs.
- Expect problems to occur, and be prepared to provide a fast/temporary fix, while an enduring solution is developed.

10.3 Operational

- Take notice of end-user views and needs.
- Never underestimate the culture change. Expect user resistance to the new system as perceptions at local level never initially align with national requirements. Never rush from pilot to national roll-out without first understanding all operational and procedural issues.
- Find a local manager to champion the system which will help smooth the implementation.
- Communicate to all what is happening.

11. Conclusions

The work management programme has succeeded in designing, developing, delivering and deploying a key operational support system for the company. The original objectives have been met, indeed WMS is now vital to a significant part of the company's operations. Where WMS has been deployed, both productivity and quality of service targets have been met.

Acknowledgements

The author gratefully acknowledges the many hundreds of people who have helped to make the programme a company success — the users in every zone for their tolerance, perseverance and suggestions for change, those in the centre who co-ordinated the requirements, drove the deployment implementation and managed on behalf of the company the overall programme, and not least those development colleagues whose dedication and technical ingenuity brought Work Manager from a laboratory model to a national production system.

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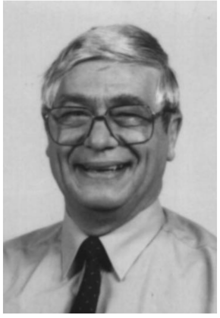

Gerry Garwood joined BT in 1960 as an apprentice. He then undertook technician duties covering, test desk, linesman, and exchange and repeater station maintenance. After graduation in 1969, he spent 4 years in computer development within the postal mechanisation department. He moved to the research department in 1973 and became involved in software systems which demonstrated that software technology could be applied to the work management domain. He was responsible for the development of the AI demonstrators, DCSS, NOMS2 and WMS. As the delivery programme manager he was responsible for further development of WMS and the deployment of it nationally. He graduated with an honours degree in electrical and electronic engineering from Brunel University, holds a Diploma in Management Studies, is a Chartered Engineer and a member of the Institution of Electrical Engineers.